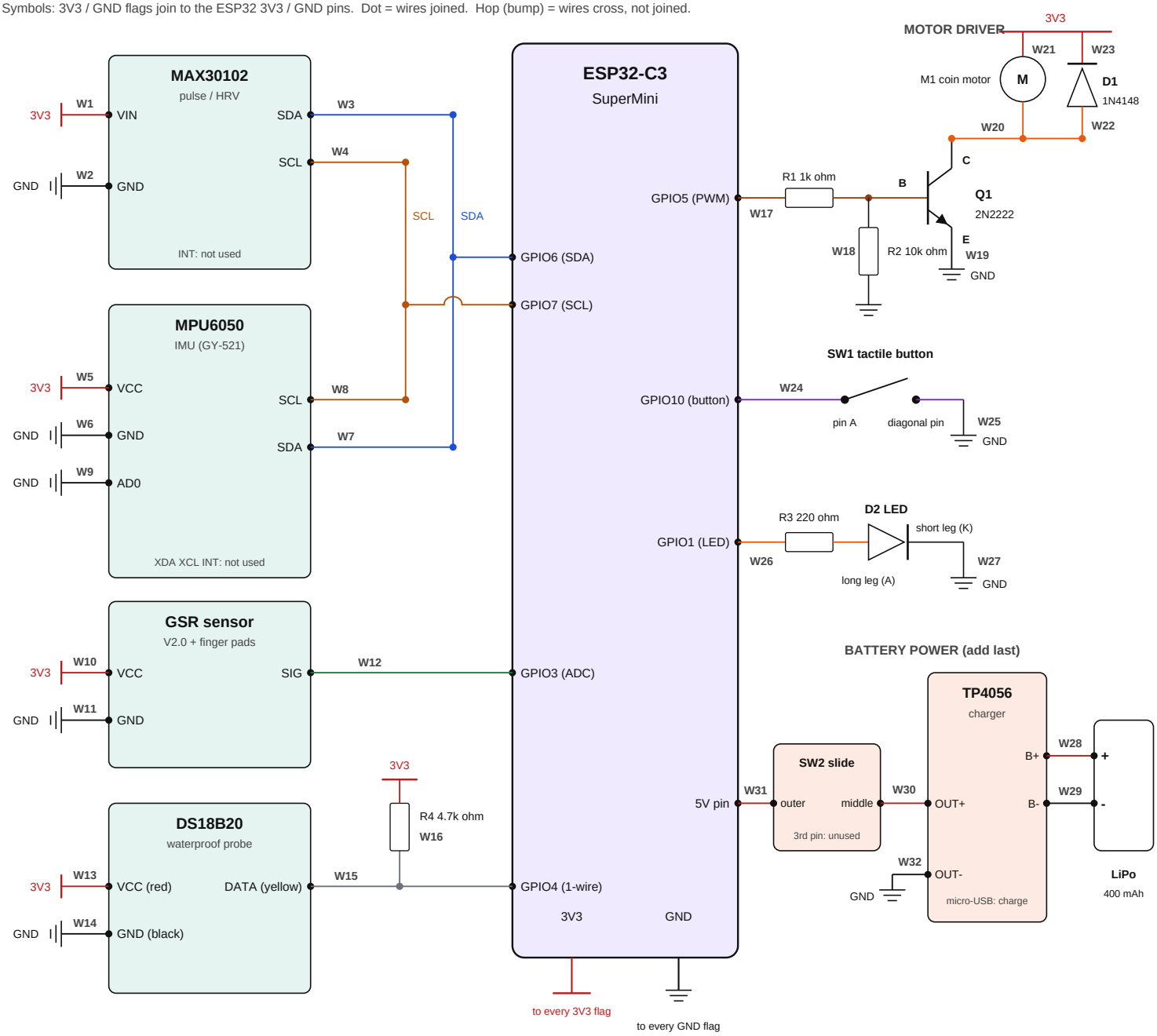
AXIO ALERT | Prototype wiring diagram

ESP32-C3 SuperMini, 3.3 V logic. Every pin has its own numbered wire (W1-W32); the table below lists them all.

Symbols: 3V3 / GND flags join to the ESP32 3V3 / GND pins. Dot = wires joined. Hop (bump) = wires cross, not joined.



WIRE LIST (W numbers match the diagram; colours are suggestions for jumper wires)

W	From	To	Colour
W1	MAX30102 VIN	3V3	red
W2	MAX30102 GND	GND	black
W3	MAX30102 SDA	ESP GPIO6	blue
W4	MAX30102 SCL	ESP GPIO7	yellow
W5	MPU6050 VCC	3V3	red
W6	MPU6050 GND	GND	black
W7	MPU6050 SDA	ESP GPIO6	blue
W8	MPU6050 SCL	ESP GPIO7	yellow
W9	MPU6050 AD0	GND	black
W10	GSR VCC	3V3	red
W11	GSR GND	GND	black
W12	GSR SIG	ESP GPIO3	green
W13	DS18B20 VCC (red wire)	3V3	red
W14	DS18B20 GND (black wire)	GND	black
W15	DS18B20 DATA (yellow wire)	ESP GPIO4	white
W16	R4 4.7k: DATA side	R4 other end: 3V3	-

W	From	To	Colour
W17	ESP GPIO5	R1 1k -> Q1 Base	brown
W18	R2 10k: Q1 Base	GND	black
W19	Q1 2N2222 Emitter	GND	black
W20	Q1 2N2222 Collector	Motor (-) lead	brown
W21	Motor (+) lead	3V3	red
W22	D1 1N4148 anode	Q1 Collector	-
W23	D1 1N4148 cathode (band)	3V3	-
W24	ESP GPIO10	Button pin A	purple
W25	Button diagonal pin	GND	black
W26	ESP GPIO1	R3 220 -> LED long leg	orange
W27	LED short leg	GND	black
W28	LiPo red	TP4056 B+	red
W29	LiPo black	TP4056 B-	black
W30	TP4056 OUT+	SW2 middle pin	red
W31	SW2 outer pin	ESP 5V pin	red
W32	TP4056 OUT-	GND (ESP GND pin)	black

NOTES

Build order: test the sensors on USB power first (skip W28-W32), then add the battery.

SAFETY: keep SW2 OFF whenever USB is plugged into the ESP32-C3, so the 5V pin is never fed from two sources.

Check the LiPo polarity with a multimeter before connecting it. Set the TP4056 charge current to about 0.4 A for the 400 mAh cell (swap the 1.2k programming resistor for about 3k).

2N2222 (TO-92): the usual pin order is E-B-C with the flat face towards you. Confirm with your datasheet or a multimeter diode test.

Tactile button: use diagonal pins (two pins on the same side are joined inside). Power every module from 3.3 V and check each board's silkscreen for pin names.

Leave unused pins open: MAX30102 INT, MPU6050 XDA / XCL / INT. If you fit a DRV2605L instead of Q1, R1, R2, D1: VIN-3V3, GND-GND, SDA-GPIO6, SCL-GPIO7, motor pads to the coin motor.

Firmware pins: Wire.begin(6, 7); analogRead(3); OneWire on 4; motor PWM on 5; button on 10 (INPUT_PULLUP); LED on 1.